

Skills Summary

Making and Qualifying Predictions from Scatter Plots

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a scatter plot

Each dot is one individual, placed by its two measurements: across for the x variable, up for the y variable. To read a dot, drop straight down to the x -axis and straight across to the y -axis. The **direction** of the cloud is what a prediction will follow: rising to the right is a positive association, falling is negative, no tilt means no association.

Example: A plot shows practice hours against free throws made for six players, with points at (1, 6), (2, 8), (3, 9), (4, 13), (5, 14) and (6, 18). How many free throws did the player who practised 4 hours make?

Step 1. The question asks for one player's actual result, not a prediction, so I read the dot itself rather than any line drawn through the cloud.

Step 2. Go across to 4 on the practice axis, up to the dot, then across to the free-throw axis.
 $\Rightarrow$ **13** free throws. The cloud rises to the right, so the association is positive.

Try it: From the same plot, how many free throws did the player who practised 2 hours make?

8 :pech

Skill 2 — Predicting with the trend line

The trend line summarises the cloud with one straight line, written $y = mx + b$. To predict, substitute the x you were given and work out y . The answer is a *typical* value for that x , not a guarantee: real points sit above and below the line, and how far they sit from it is how far your prediction may be out.

Example: The trend line for that plot is $y = 2.5x + 3.5$. Predict the free throws made after 5 hours of practice.

Step 1. A typical value is wanted rather than one player's result, so this is a job for the line and not for the dots.

Step 2. Substitute $x = 5$: $2.5 \times 5 = 12.5$, and $12.5 + 3.5 = 16$.

Step 3. The player who did practise 5 hours actually made 14, so a prediction here is good to about two throws.

$\Rightarrow$ **16** free throws

Try it: Use $y = 2.5x + 3.5$ to predict the free throws made after 2 hours of practice.

5.8 :check

Skill 3 — Deciding whether to trust a prediction

Three questions to ask of every number the line produces:

- Is the x *inside* the range that was measured? Outside it you are extrapolating, and nothing in the data says the pattern continues.
- Is the answer *possible*? A score above the maximum mark, or a negative time, means the model has been pushed past where it applies.
- Is there really a pattern? A cloud with no tilt supports no prediction at all — and even a strong pattern shows association, never cause.

Example: Use $y = 2.5x + 3.5$ to predict the free throws made after 40 hours of practice, then judge the answer.

Step 1. The line will happily return a number for any input, so before trusting it I check the input against the range of practice hours that were actually measured, namely 1 to 6.

Step 2. Substitute $x = 40$: $2.5 \times 40 = 100$, and $100 + 3.5 = 103.5$.

Step 3. That x is far outside the measured range, and no player attempts more than a few dozen throws, so the number is arithmetic, not evidence.

⇒ **103.5** — calculated correctly, and not a prediction anyone should use.

Try it: What does $y = 2.5x + 3.5$ give for 30 hours of practice, and would you trust it?

check: **78.5** — far outside the measured range